

Sharp endpoint regularity for rank-one compact Herz–Schur multipliers

Literature-implied answer note for arXiv:2305.09291

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Status. Literature-implied answer (endpoint subcase). This note completely answers the question at $p = \infty$. It does not settle the finite- p cases.

Source question

Let (G, K) be a compact symmetric Gelfand pair of rank one and let $d = \dim(G/K)$. Dumas proves regularity for every K -bi-invariant $S_p(L^2(G))$ -multiplier by testing it against an S_p -valued operator path T . Remark 3.18 of [1], on PDF page 28, asks whether this regularity is optimal, observing that the paper does not construct multipliers attaining the threshold.

Set

$$a = \frac{d-1}{2}, \quad \mathcal{R}(a) = ([a], a - [a]).$$

Regularity pairs are ordered lexicographically, as in [1].

Identification and result

The missing endpoint observation is that compactness forces every S_∞ Herz–Schur multiplier to be a unitary matrix coefficient. Indeed, Spronk’s Corollary 5.4 identifies invariant measurable Schur multipliers on an amenable locally compact group with its Fourier–Stieltjes algebra $B(G)$ [2]. Compact groups are amenable, and elements of $B(G)$ are precisely coefficients of unitary representations.

Theorem 1 (Sharp $p = \infty$ endpoint). *Let (G, K) be a compact symmetric Gelfand pair of rank one. Every K -bi-invariant $S_\infty(L^2(G))$ -multiplier φ satisfies*

$$\varphi \circ \exp \in C^{\mathcal{R}(a)}(Q).$$

Moreover, for every regularity pair $(r, \delta) > \mathcal{R}(a)$ there is a K -bi-invariant S_∞ -multiplier $\varphi_{r,\delta}$ such that

$$\varphi_{r,\delta} \circ \exp \notin C^{(r,\delta)}(Q).$$

Thus $\mathcal{R}(a)$ is the optimal endpoint regularity.

Proof. If φ is an S_∞ -multiplier, its Toeplitz kernel

$$u(s, t) = \varphi(st^{-1})$$

defines an invariant measurable Schur multiplier on $B(L^2(G))$. Since G is compact and hence amenable, Spronk's Corollary 5.4 gives $\varphi \in B(G)$. Therefore there are a unitary representation π of G and vectors ξ, η such that $\varphi(g) = \langle \pi(g)\xi, \eta \rangle$. The function is already assumed K -bi-invariant, so Dumas's Theorem A applies and yields $\varphi \circ \exp \in C^{\mathcal{R}(a)}(Q)$.

For sharpness, Dumas's Corollary 3.17(5) states that for every $(r, \delta) > \mathcal{R}(a)$ there is a K -bi-invariant unitary matrix coefficient ψ which does not belong to $C^{(r, \delta)}(Q)$. Such a coefficient is an S_∞ -multiplier: if $\psi(g) = \langle \pi(g)\xi, \eta \rangle$, then its Toeplitz kernel factors as

$$\psi(st^{-1}) = \langle \pi(t^{-1})\xi, \pi(s^{-1})\eta \rangle.$$

The two vector-valued factors are bounded, so the standard Schur factorization gives $\|S_\psi\| \leq \|\xi\| \|\eta\|$. Hence ψ is the required S_∞ -multiplier. $\square$

Integer endpoint consequence

If $a \notin \mathbb{Z}$, the theorem gives exactly the regularity of the operator path T used in Corollary 3.13 of [1]. If $a = m \in \mathbb{Z}$, that path is proved only to lie in $C^{(m-1, 1)}$, whereas the theorem shows that every S_∞ -multiplier lies in $C^{(m, 0)} = C^m$, and that C^m is sharp. Thus in the integer case the path regularity is not the true multiplier threshold; no endpoint multiplier can have only the weaker regularity of T .

Provenance, search, and limitations

The implication above is not presented as new mathematics. It combines Dumas's sharp rank-one matrix-coefficient theorem and optimality corollary with Spronk's standard amenable-group identification. Spronk predates the source question and does not claim to answer it; the connection is the identification made in this packet. Exact-phrases and title searches of arXiv and the published source located no separate paper explicitly resolving Remark 3.18. Nothing here addresses optimality for finite p .

References

- [1] G. Dumas, *Regularity of matrix coefficients of a compact symmetric pair of Lie groups*, Trans. Amer. Math. Soc. 377 (2024), 7421–7474; arXiv:2305.09291.
- [2] N. Spronk, *Measurable Schur multipliers and completely bounded multipliers of the Fourier algebras*, Proc. London Math. Soc. (3) 89 (2004), 161–192; arXiv:math/0210304. See Corollary 5.4.